12 percent of GDP in 2023. This implies that without active policy measures aimed at reducing the fiscal deficit, debt sustainability would be jeopardized.

5. Stress tests suggest that real GDP growth shocks have the largest impact on debt indicators. An adverse shock to growth,[2] whereby real GDP contracts by almost 2 percent in 2019–20 in each year, causes public debt to increase to 82 percent of GDP, and public gross financing needs exceed 21 percent of GDP by 2020. This scenario, however, is based on historical GDP growth which, as noted above, has been characterized by high volatility, and the realization of shocks of such magnitude is relatively unlikely. Stress tests based on standardized primary balance, real interest and exchange rate shocks[3] reveal a moderate deterioration relative to the baseline, with real exchange rate depreciation having a somewhat larger impact.

[2] The shock assumes that real GDP growth is reduced by 1 standard deviation of real GDP growth rates over the past 10 years.

[3] The shock assumes the maximum historical movement of the REER over the past 10 years.

©International Monetary Fund. Not for Redistribution

Figure A1. Armenia Public Sector Debt Sustainability Analysis (DSA) - Baseline Scenario

(in percent of GDP unless otherwise indicated)

Debt, Economic and Market Indicators [1/]

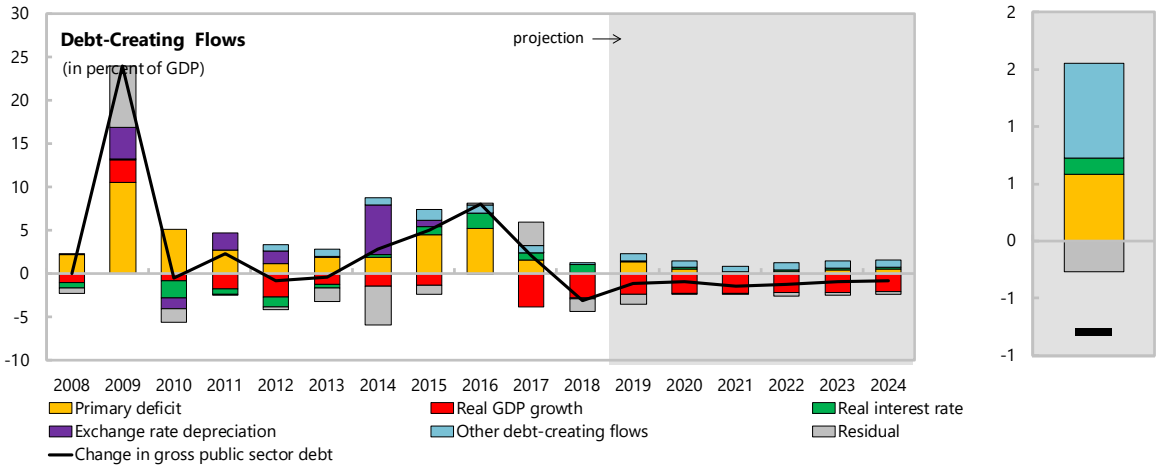
	Actual			Projections					
	2008-2016 [2/]	2017	2018	2019	2020	2021	2022	2023	2024
Nominal gross public debt	41.0	58.9	55.8	54.6	53.7	52.3	51.1	50.1	49.3
Public gross financing needs	7.9	6.6	6.1	7.5	10.2	6.4	6.4	6.2	6.1
Net public debt		52.1	49.7	48.7	48.4	47.7	46.8	46.2	45.8
Real GDP growth (in percent)	1.9	7.5	5.2	4.6	4.5	4.5	4.5	4.5	4.5
Inflation (GDP deflator, in percent)	3.7	2.2	2.5	4.3	3.8	4.4	4.2	4.0	3.9
Nominal GDP growth (in percent)	5.7	9.9	7.8	9.1	8.5	9.1	8.9	8.7	8.6
Effective interest rate (in percent) [4/]	2.9	4.2	4.2	4.7	4.4	5.1	5.0	4.7	4.4

As of March 31, 2019

Sovereign Spreads		
Spread (bp) 3/		369
CDS (bp)		n.a.
Ratings	**Foreign**	**Local**
Moody's	B1	n.a.
S&Ps	BB-	n.a.
Fitch	B+	n.a.

Contribution to Changes in Public Debt

	Actual			Projections							
	2008-2016	2017	2018	2019	2020	2021	2022	2023	2024	cumulative	debt-stabilizing primary balance [9/]
Change in gross public sector debt	4.5	2.1	-3.1	-1.1	-0.9	-1.4	-1.3	-0.9	-0.8	-6.4	
Identified debt-creating flows	4.8	-0.6	-1.6	1.1	0.2	-0.4	0.1	0.3	0.4	1.7	
Primary deficit	3.9	1.5	0.1	1.4	0.6	0.0	0.2	0.4	0.6	3.2	-0.2
Primary (noninterest) revenue and grants	21.4	21.2	21.7	22.6	22.8	22.8	22.9	23.2	23.1	137.4	
Primary (noninterest) expenditure	25.3	22.8	21.8	24.0	23.3	22.8	23.1	23.6	23.7	140.6	
Automatic debt dynamics [5/]	0.3	-2.9	-2.0	-1.2	-1.0	-1.0	-0.9	-1.0	-1.0	-6.1	
Interest rate/growth differential [6/]	-1.0	-3.0	-1.9	-2.2	-2.0	-2.0	-1.9	-1.9	-1.9	-12.0	
Of which: real interest rate	-0.2	0.9	0.9	0.1	0.2	0.2	0.3	0.2	0.1	1.2	
Of which: real GDP growth	-0.9	-3.9	-2.8	-2.3	-2.3	-2.2	-2.2	-2.1	-2.1	-13.2	
Exchange rate depreciation [7/]	1.4	0.0	0.0	...	...	...	...	...	...	...	
Other identified debt-creating flows	0.5	0.8	0.2	0.9	0.7	0.6	0.8	0.8	0.8	4.6	
Domestic net lend./drawdown of gov. dep.	0.0	0.0	-0.5	0.0	-0.2	-0.3	0.0	0.0	0.0	-0.4	
Contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
External net lending	0.5	0.8	0.8	0.9	0.9	0.9	0.8	0.8	0.8	5.1	
Residual, including asset changes [8/]	-0.3	2.8	-1.5	-1.1	-0.1	-0.1	-0.4	-0.3	-0.3	-2.3	



Source: IMF staff.

1/ Public sector is defined as general government.

2/ Based on available data.

3/ Bond Spread over U.S. Bonds.

4/ Defined as interest payments divided by debt stock at the end of previous year.

5/ Derived as [(r - p(1+g) - g + ae(1+r)]/(1+g+p+gp)) times previous period debt ratio, with r = interest rate; p = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

6/ The real interest rate contribution is derived from the denominator in footnote 4 as r - π (1+g) and the real growth contribution as -g.

7/ The exchange rate contribution is derived from the numerator in footnote 2/ as ae(1+r).

8/ Includes asset changes and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

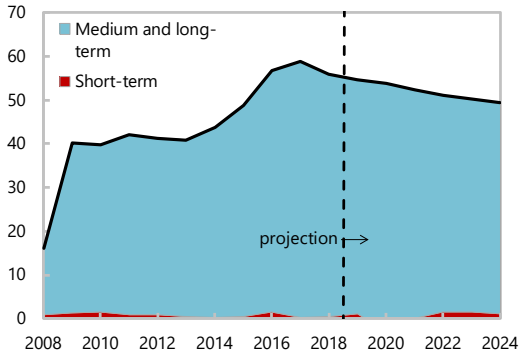
9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

©International Monetary Fund. Not for Redistribution

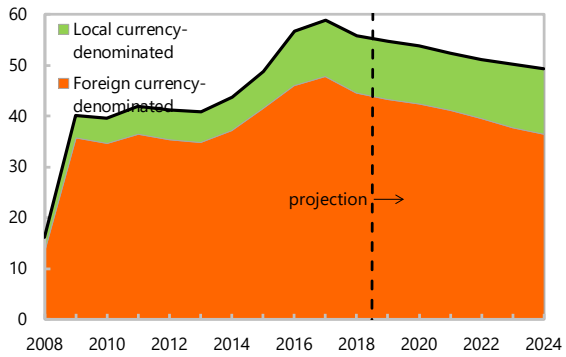
Figure A2. Armenia Public DSA - Composition of Public Debt and Alternative Scenarios

Composition of Public Debt

By Maturity
(in percent of GDP)



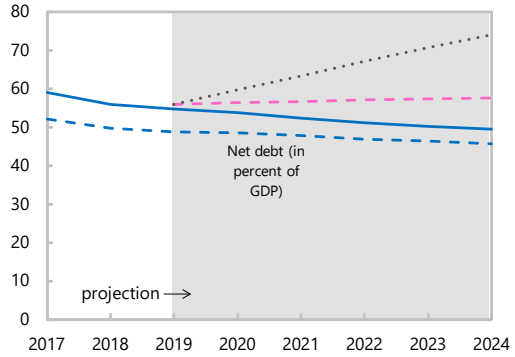
By Currency
(in percent of GDP)



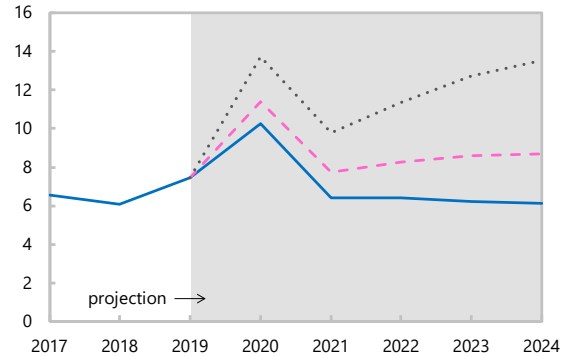
Alternative Scenarios



Gross Nominal Public Debt
(in percent of GDP)



Public Gross Financing Needs
(in percent of GDP)



Underlying Assumptions
(in percent)

Baseline Scenario	2019	2020	2021	2022	2023	2024
Real GDP growth	4.6	4.5	4.5	4.5	4.5	4.5
Inflation	4.3	3.8	4.4	4.2	4.0	3.9
Primary Balance	-1.4	-0.6	0.0	-0.2	-0.4	-0.6
Effective interest rate	4.7	4.4	5.1	5.0	4.7	4.4
Constant Primary Balance Scenario						
Real GDP growth	4.6	4.5	4.5	4.5	4.5	4.5
Inflation	4.3	3.8	4.4	4.2	4.0	3.9
Primary Balance	-1.4	-1.4	-1.4	-1.4	-1.4	-1.4
Effective interest rate	4.7	4.4	5.1	5.1	4.8	4.6

Historical Scenario	2019	2020	2021	2022	2023	2024
Real GDP growth	4.6	2.3	2.3	2.3	2.3	2.3
Inflation	4.3	3.8	4.4	4.2	4.0	3.9
Primary Balance	-1.4	-3.5	-3.5	-3.5	-3.5	-3.5
Effective interest rate	4.7	4.4	5.0	5.1	4.7	4.5

Source: IMF staff.

©International Monetary Fund. Not for Redistribution